

Topics: 14.3 Partial Derivatives (Part II); 14.4 Tangent Planes and Linear Approximations; 14.5 The Chain Rule

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1. For the function $f(x, y) = x^4y^2 - x^3y$, find $f_{xxx}(x, y)$ and $f_{xyx}(x, y)$.

Taking partial derivatives with respect to x ,

$$f_x = 4x^3y^2 - 3x^2y, \quad f_{xx} = 12x^2y^2 - 6xy, \quad f_{xxx} = 24xy^2 - 6y.$$

Taking partial derivatives of f_x with respect to y and then x ,

$$f_{xy} = \frac{\partial}{\partial y}(f_x) = 8x^3y - 3x^2, \quad f_{xyx} = \frac{\partial}{\partial x}(f_{xy}) = 24x^2y - 6x.$$

2. Use Clairaut's theorem to show that there is no function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = e^x \cos y \quad \text{and} \quad \frac{\partial f}{\partial y} = e^x \sin y.$$

If such a function f were to exist, then by differentiating, we would have

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = -e^x \sin y \quad \text{and} \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = e^x \sin y.$$

Thus, such an f would have continuous second order partial derivatives with the property that

$$\frac{\partial^2 f}{\partial y \partial x} \neq \frac{\partial^2 f}{\partial x \partial y}.$$

However, this would violate Clairaut's theorem, so no such function $f(x, y)$ can exist.

3. Find the requested partial derivative, where z is defined implicitly by the given equation.

- (a) Find $\frac{\partial z}{\partial x}$ given that $x^2 \sin(y + z) = x + yz$.

Taking the partial derivative of both sides with respect to x ,

$$2x \sin(y + z) + x^2 \cos(y + z) \frac{\partial z}{\partial x} = 1 + y \frac{\partial z}{\partial x}.$$

Rearranging,

$$x^2 \cos(y + z) \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial x} = 1 - 2x \sin(y + z).$$

Factoring the left-hand side,

$$(x^2 \cos(y + z) - y) \frac{\partial z}{\partial x} = 1 - 2x \sin(y + z).$$

Finally, we divide through to get

$$\frac{\partial z}{\partial x} = \frac{1 - 2x \sin(y + z)}{x^2 \cos(y + z) - y}.$$

- (b) Find $\frac{\partial z}{\partial y}$ given that $\tan(x + y + z) = xyz$.

Taking the partial derivative of both sides with respect to y ,

$$\sec^2(x + y + z) \left(1 + \frac{\partial z}{\partial y} \right) = xz + xy \frac{\partial z}{\partial y}.$$

Expanding, rearranging, and factoring:

$$\sec^2(x + y + z) + \sec^2(x + y + z) \frac{\partial z}{\partial y} = xz + xy \frac{\partial z}{\partial y},$$

$$\sec^2(x + y + z) \frac{\partial z}{\partial y} - xy \frac{\partial z}{\partial y} = xz - \sec^2(x + y + z).$$

$$(\sec^2(x + y + z) - xy) \frac{\partial z}{\partial y} = xz - \sec^2(x + y + z).$$

Finally, dividing through, we get

$$\frac{\partial z}{\partial y} = \frac{xz - \sec^2(x + y + z)}{\sec^2(x + y + z) - xy}.$$

4. Find an equation of the tangent plane to the surface at the given point.

- (a) $z = 4x^2 + y^2 - 9y$ at the point on the surface with $(x, y) = (1, 4)$.

We write $z = f(x, y)$ where $f(x, y) = 4x^2 + y^2 - 9y$.

We first calculate that $f(1, 4) = -16$.

Now compute the partial derivatives:

$$f_x(x, y) = 8x, \quad f_y(x, y) = 2y - 9.$$

Evaluating at $(1, 4)$, we find that $f_x(1, 4) = 8$ and $f_y(1, 4) = -1$.

Thus, an equation of the tangent plane is

$$z + 16 = 8(x - 1) - 1(y - 4).$$

This can be written as

$$z = 8x - y - 20.$$

- (b) $z = y \ln(x)$ at the point on the surface with $(x, y) = (e, 6)$.

We write $z = f(x, y)$ where $f(x, y) = y \ln(x)$.

We first calculate that $f(e, 6) = 6$.

Now compute the partial derivatives:

$$f_x(x, y) = \frac{y}{x}, \quad f_y(x, y) = \ln(x).$$

Evaluating at $(e, 6)$, we find that $f_x(e, 6) = \frac{6}{e}$ and $f_y(e, 6) = 1$.

Thus, an equation of the tangent plane is

$$z - 6 = \frac{6}{e}(x - e) + (y - 6).$$

This can be written as

$$z = \frac{6}{e}x + y - 6.$$

5. Estimate the value using a linear approximation. *Your final answer should be a decimal number.*

(a) Given $f(x, y) = \frac{5\sqrt{y}}{x}$, estimate $f(-0.99, 1.01)$ using the point $(-1, 1)$.

We use the linearization formula

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

with $(a, b) = (-1, 1)$. We first compute that $f(-1, 1) = -5$.

Now compute the partial derivatives:

$$f_x(x, y) = -\frac{5\sqrt{y}}{x^2}, \quad f_y(x, y) = \frac{5}{2x\sqrt{y}}.$$

Evaluating at $(-1, 1)$, we find that $f_x(-1, 1) = -5$ and $f_y(-1, 1) = -\frac{5}{2}$.

Thus, the linearization of f at $(-1, 1)$ is

$$L(x, y) = -5 - 5(x + 1) - \frac{5}{2}(y - 1).$$

Substituting in $(x, y) = (-0.99, 1.01)$ gives

$$f(-0.99, 1.01) \approx L(-0.99, 1.01) = -5 - 5(0.01) - \frac{5}{2}(0.01) = -5.075.$$

(b) Given $g(x, y, z) = y \arctan(x^2 + z)$, estimate $g(0.9, 2.1, -1.1)$ using the point $(1, 2, -1)$.

We use the linearization formula

$$L(x, y, z) = g(a, b, c) + g_x(a, b, c)(x - a) + g_y(a, b, c)(y - b) + g_z(a, b, c)(z - c)$$

with $(a, b, c) = (1, 2, -1)$. We first compute that

$$g(1, 2, -1) = 2 \arctan(1^2 - 1) = 2 \arctan(0) = 0.$$

Now compute the partial derivatives:

$$g_x(x, y, z) = \frac{2xy}{1 + (x^2 + z)^2}, \quad g_y(x, y, z) = \arctan(x^2 + z), \quad g_z(x, y, z) = \frac{y}{1 + (x^2 + z)^2}.$$

Evaluating at $(1, 2, -1)$, we find that

$$g_x(1, 2, -1) = 4, \quad g_y(1, 2, -1) = 0, \quad g_z(1, 2, -1) = 2.$$

Thus, the linearization of g at $(1, 2, -1)$ is

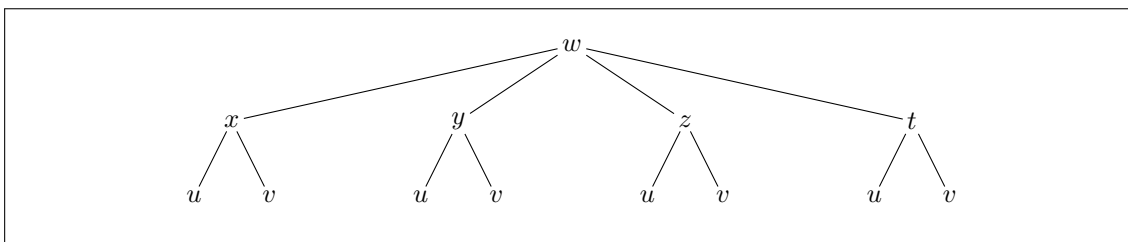
$$L(x, y, z) = 4(x - 1) + 0(y - 2) + 2(z + 1).$$

Substituting in $(x, y, z) = (0.9, 2.1, -1.1)$ gives

$$g(0.9, 2.1, -1.1) \approx L(0.9, 2.1, -1.1) = 4(0.9 - 1) + 2(-1.1 + 1) = -0.6.$$

6. Suppose that $w = f(x, y, z, t)$, where $x = x(u, v)$, $y = y(u, v)$, $z = z(u, v)$, and $t = t(u, v)$.

(a) Draw a tree diagram showing the dependence of w on u and v .



(b) Use the tree diagram to write expressions for $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$.

$$\begin{aligned} \frac{\partial w}{\partial u} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial u} + \frac{\partial w}{\partial t} \frac{\partial t}{\partial u} \\ \frac{\partial w}{\partial v} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial v} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial v} + \frac{\partial w}{\partial t} \frac{\partial t}{\partial v} \end{aligned}$$

7. Use the Chain Rule to find the indicated derivatives. Your final answer may be written in terms of any variables appearing in the problem statement. Do not simplify your final answer.

(a) Find $\frac{dz}{dt}$ given that $z = \frac{x - y}{x + 2y}$, $x = e^{\pi t}$, and $y = e^{-\pi t}$.

We first compute

$$\frac{dx}{dt} = \pi e^{\pi t} \quad \text{and} \quad \frac{dy}{dt} = -\pi e^{-\pi t}.$$

Next,

$$\frac{\partial z}{\partial x} = \frac{(x + 2y)(1) - (x - y)(1)}{(x + 2y)^2} = \frac{3y}{(x + 2y)^2},$$

and

$$\frac{\partial z}{\partial y} = \frac{(x + 2y)(-1) - (x - y)(2)}{(x + 2y)^2} = \frac{-3x}{(x + 2y)^2}.$$

By the Chain Rule,

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = \left(\frac{3y}{(x + 2y)^2} \right) \pi e^{\pi t} + \left(\frac{-3x}{(x + 2y)^2} \right) (-\pi e^{-\pi t}).$$

- (b) Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ given that $z = \arctan(x^2 + y^2)$, $x = s \ln t$, and $y = te^s$.

First compute

$$\frac{\partial x}{\partial s} = \ln t, \quad \frac{\partial x}{\partial t} = \frac{s}{t}, \quad \frac{\partial y}{\partial s} = te^s, \quad \frac{\partial y}{\partial t} = e^s.$$

Also,

$$\frac{\partial z}{\partial x} = \frac{2x}{1 + (x^2 + y^2)^2}, \quad \frac{\partial z}{\partial y} = \frac{2y}{1 + (x^2 + y^2)^2}.$$

By the Chain Rule,

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = \left(\frac{2x}{1 + (x^2 + y^2)^2} \right) (\ln t) + \left(\frac{2y}{1 + (x^2 + y^2)^2} \right) (te^s).$$

Similarly,

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} = \left(\frac{2x}{1 + (x^2 + y^2)^2} \right) \left(\frac{s}{t} \right) + \left(\frac{2y}{1 + (x^2 + y^2)^2} \right) (e^s).$$

8. Suppose f is differentiable and $x = g(t)$, $y = h(t)$. Given that

$$g(2) = 4, \quad g'(2) = -3, \quad h(2) = 5, \quad h'(2) = 6, \quad f_x(4, 5) = 2, \quad f_y(4, 5) = 8,$$

determine the derivative of $f(x, y)$ with respect to t at $t = 2$.

By the Chain Rule,

$$\frac{d}{dt} f(x, y) = f_x(x, y) \frac{dx}{dt} + f_y(x, y) \frac{dy}{dt}.$$

Since $x = g(t)$ and $y = h(t)$, this becomes

$$\frac{d}{dt} f(g(t), h(t)) = f_x(g(t), h(t)) g'(t) + f_y(g(t), h(t)) h'(t).$$

Evaluating at $t = 2$, we get

$$\begin{aligned} \left. \frac{d}{dt} f(g(t), h(t)) \right|_{t=2} &= f_x(g(2), h(2)) g'(2) + f_y(g(2), h(2)) h'(2) \\ &= f_x(4, 5)(-3) + f_y(4, 5)(6) \\ &= 2(-3) + 8(6) \\ &= -6 + 48 \\ &= 42. \end{aligned}$$

Therefore, the derivative at $t = 2$ is 42.

9. Let $S(x, y)$ represent the thickness (in millimeters) of a coating on a flat metal panel at the point (x, y) . A sensor moves across the panel along the path given parametrically by

$$x(t) = \sqrt{1+t}, \quad y(t) = 3 + t^2,$$

where x and y are measured in centimeters and t is measured in seconds. Suppose that

$$S_x(2, 12) = 4 \quad \text{and} \quad S_y(2, 12) = 2.$$

Determine the rate at which the coating thickness is changing along the sensor's path at $t = 3$.

By the Chain Rule,

$$\frac{dS}{dt} = S_x(x, y) \frac{dx}{dt} + S_y(x, y) \frac{dy}{dt}.$$

We compute

$$\frac{dx}{dt} = \frac{1}{2\sqrt{1+t}} \quad \text{and} \quad \frac{dy}{dt} = 2t.$$

At $t = 3$, we find that $x(3) = \sqrt{4} = 2$, $y(3) = 3 + 9 = 12$, $\frac{dx}{dt}\big|_{t=3} = \frac{1}{4}$, and $\frac{dy}{dt}\big|_{t=3} = 6$.

Thus, by the Chain Rule,

$$\frac{dS}{dt}\bigg|_{t=3} = S_x(2, 12) \left(\frac{1}{4}\right) + S_y(2, 12)(6) = 4 \left(\frac{1}{4}\right) + 2(6) = 1 + 12 = 13.$$

Therefore, the coating thickness is increasing at a rate of 13 millimeters per second.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Clairaut's Theorem. If f is a function of two variables for which the second order partial derivatives f_{xy} and f_{yx} are continuous on a disk that contains a point (a, b) , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Tangent Plane. Let f be a function of two variables with continuous first partial derivatives, and let $z_0 = f(x_0, y_0)$. The tangent plane to $z = f(x, y)$ at (x_0, y_0, z_0) has equation

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Linearization. Let f be a function of two variables whose first partial derivatives are continuous. The linearization of f at (x_0, y_0) is the linear function

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

For (x, y) near (x_0, y_0) , we have the approximation $f(x, y) \approx L(x, y)$.

Similarly, if f is a function of three variables with continuous first partial derivatives, then

$$L(x, y, z) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0)$$

is the linearization of f at (x_0, y_0, z_0) , and $f(x, y, z) \approx L(x, y, z)$ for (x, y, z) near (x_0, y_0, z_0) .

The Chain Rule. Suppose that u is a differentiable function of the n variables $x_1, x_2, \dots, x_n$, and that each x_j is, in turn, a differentiable function of the m variables $t_1, t_2, \dots, t_m$. Then u is a differentiable function of $t_1, t_2, \dots, t_m$, and for each $i = 1, 2, \dots, m$,

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \cdots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i}.$$

Suggested Textbook Problems

Section 14.3: 1-74

Section 14.4: 1-46

Section 14.5: 1-55